Light

GCSE Computer Science — Data Representation: Numbers (Set 2)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A computer stores all data using the binary number system.
What is the base of the binary number system?

- **A.** Base-2
- **B.** Base-8
- **C.** Base-10
- **D.** Base-16

Q2 (1 mark)

In an 8-bit binary number, what is the place value of the rightmost bit?

- **A.** 0

- B. 1
- C. 2
- D. 8

Q3 (1 mark)

What is the denary number 9 in binary?

- A. 1000
- B. 1010
- C. 1001
- D. 1011

Q4 (1 mark)

What is the binary number 1010 in denary?

- A. 8
- B. 9
- C. 11
- D. 10

Q5 (1 mark)

Hexadecimal uses digits 0-9 and letters A-F.

What denary value does the letter C represent?

- ○ **A.** 10
- ○ **B.** 11
- ○ **C.** 13
- ○ **D.** 12

Q6 (1 mark)

Each hexadecimal digit can be represented by a group of binary digits. How many binary bits are needed to represent one hexadecimal digit?

- ○ **A.** 2
- ○ **B.** 4
- ○ **C.** 8
- ○ **D.** 16

Q7 (1 mark)

How many different values can be represented using 3 bits?

- ○ **A.** 3
- ○ **B.** 6
- ○ **C.** 9
- ○ **D.** 8

Q8 (1 mark)

What is the result of adding the binary numbers 0101 and 0010?

- ○ **A.** 0110
- ○ **B.** 0111
- ○ **C.** 1000
- ○ **D.** 1001

Q9 (1 mark)

The binary number 00000101 is shifted one place to the left.
What is the result?

- ○ **A.** 00001010
- ○ **B.** 00000010
- ○ **C.** 00000110
- ○ **D.** 00010100

Q10 (1 mark)

The binary number 00010000 is shifted one place to the right.
What is the result?

- ○ **A.** 00100000
- ○ **B.** 00001000
- ○ **C.** 00010001

- ○ **D.** 00000100

Q11 (1 mark)

An unsigned 8-bit binary number has a minimum and maximum value.
What is the minimum value that can be stored?

- ○ **A.** -128
- ○ **B.** -1
- ○ **C.** 0
- ○ **D.** 1

Q12 (1 mark)

In two's complement representation, the leftmost bit has a special meaning.
What does the leftmost bit indicate?

- ○ **A.** Whether the number is odd or even
- ○ **B.** Whether the number is positive or negative
- ○ **C.** The number of bits used
- ○ **D.** Whether overflow has occurred

Q13 (2 marks)

Convert the denary number 156 to an 8-bit binary number.

- A. 10011100
- B. 10011010
- C. 10011110
- D. 10011000

Q14 (2 marks)

Convert the 8-bit binary number 11100011 to denary.

- A. 223
- B. 225
- C. 227
- D. 229

Q15 (2 marks)

Convert the binary number 10101100 to hexadecimal.

- A. AB
- B. AC
- C. CA
- D. BC

Q16 (2 marks)

Convert the hexadecimal number 7D to binary.

- ○ **A.** 01111100
- ○ **B.** 01111101
- ○ **C.** 01111110
- ○ **D.** 01111011

Q17 (2 marks)

Calculate the result of adding the binary numbers 00110011 and 00011001.

- ○ **A.** 01001010
- ○ **B.** 01001100
- ○ **C.** 01010100
- ○ **D.** 01001000

Q18 (2 marks)

The binary number 00000110 is shifted three places to the left. What denary value does the result represent?

- ○ **A.** 24
- ○ **B.** 36
- ○ **C.** 54

- **D.** 48

Q19 (2 marks)

The binary number 01010000 is shifted two places to the right.
What denary value does the result represent?

- **A.** 10
- **B.** 20
- **C.** 40
- **D.** 80

Q20 (2 marks)

A student attempts to store the result of adding 180 + 90 in an unsigned 8-bit binary register.
What problem occurs?

- **A.** No problem occurs because the result fits in 8 bits
- **B.** Overflow occurs because the result exceeds the maximum value of 255
- **C.** Underflow occurs because the result is negative
- **D.** The calculation is impossible because both numbers are too large

Q21 (2 marks)

In 8-bit two's complement, what is the binary representation of -5?

- A. 00000101
- B. 10000101
- C. 11111010
- D. 11111011

Q22 (2 marks)

Convert the hexadecimal number 5E to denary.

- A. 88
- B. 90
- C. 92
- D. 94

Q23 (2 marks)

A password system allows each character to be one of 256 possible values.
How many bits are used to store each character?

- A. 6
- B. 7
- C. 8
- D. 9

Q24 (3 marks)

Two unsigned 8-bit binary numbers are added together:
10110011 + 01101010
What is the correct denary result, and what problem occurs?

- ○ **A.** Result is 285, and overflow occurs because this exceeds 255
- ○ **B.** Result is 29, and no problem occurs
- ○ **C.** Result is 285, and no problem occurs
- ○ **D.** Result is 157, and underflow occurs

Q25 (3 marks)

Convert the denary number -73 to an 8-bit two's complement binary number.

- ○ **A.** 10110111
- ○ **B.** 10111001
- ○ **C.** 11001000
- ○ **D.** 11000111

Q26 (3 marks)

Convert the denary number 214 to hexadecimal.

- ○ **A.** D4
- ○ **B.** D6

- **C.** D8
- **D.** DA

Q27 (3 marks)

A signed 8-bit two's complement number 11101000 undergoes an arithmetic right shift of two places.
What is the result in denary?

- **A.** -6
- **B.** -12
- **C.** -24
- **D.** -48

Q28 (3 marks)

A student performs the following binary calculation:
00100101 + 00011110 + 00010101
What is the final result in denary?

- **A.** 84
- **B.** 86
- **C.** 88
- **D.** 90

Q29 (3 marks)

The 8-bit binary pattern 11010100 can be interpreted as either an unsigned integer or a signed two's complement integer.
What are the two possible denary values?

- ○ **A.** Unsigned: 212, Signed: -44
- ○ **B.** Unsigned: 212, Signed: -212
- ○ **C.** Unsigned: 44, Signed: -212
- ○ **D.** Unsigned: 44, Signed: -44

Q30 (3 marks)

A memory address is represented in hexadecimal as #B5C.
Assuming this is a 12-bit value (4 bits per hex digit), what is the equivalent value in denary?

- ○ **A.** 2876
- ○ **B.** 2892
- ○ **C.** 2908
- ○ **D.** 2924

Total: **55** marks • Time: **30 min**

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